

MR-based objective assessment of COVID-19 olfactory rehabilitation based on VBM, fMRI, DTI and ASL

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Introduction

Studies have found that over 50% of COVID-19 patients experience sudden loss of smell (anosmia), higher than other respiratory diseases, which may be triggered by different factors: olfactory cleft syndrome, local inflammation in the nasal epithelium, early apoptosis of olfactory cells, changes in olfactory cilia and odor transmission, damage to microglial cells, effect on olfactory bulbs, epithelial olfactory injury and impairment of olfactory neurons and stem cells¹. Olfactory rehabilitation is a promising approach to treatment and the Connecticut Chemosensorial Clinical Research Centre test is currently used to assess olfactory function². This work explores the use of fully synchronized olfactory fMRI and studies anatomical, connectivity and perfusion changes as an objective tool to assess the physiological activation of the olfactory system and related areas after the targeted rehabilitation therapy focused on orientating responses, divided attention, working memory, perception, denomination, executive functions, discrimination process and emotion .

Material & Methods

As part of an IRB approved study, MRI data from 4 female patients ranged 45-54y, that presented persistent olfactory loss after COVID-19 infection, was acquired on a 3T SignaTM Premier (GE Healthcare, Waukesha, USA) at Hospital Quirónsalud (Pozuelo de Alarcón, Spain), after obtaining written informed consent. Participants underwent a targeted olfactory rehabilitation therapy of 10 sessions over 2.5 months; MRI data were obtained prior to the start of the therapy and after the end of the therapy. The MRI protocol included an anatomical 3D-T1-MPRAGE (1mm iso-voxel), a CUBE T2-FLAIR, a 10 minutes fMRI study (TR=1s, TE=20.6ms, FOV=22cm, voxel size 2.3mm iso, flip angle=62o, HyperBand factor=3, ASSET factor=2) fully synchronized with the scanner trigger and respiratory belt (Fig.1) delivered two different odours (mint and vanilla) randomized in an event-related fashion within the duration of the study^{3,4}, a diffusion HARDI scheme (150 directions, b-value=1500 s/mm²), and a pcASL scheme (PLD=1.45s, labelling time=1.5s) sequences. SPM12 was used to segment the structural MPRAGE T1 images and to realign, coregister and run a first and second analysis (paired t-test) on the fMRI images. DSI Studio was used to process the diffusion-weighted images, reconstruct the DTI model, and derive the diffusion tensor scalar maps. Longitudinal changes in fractional anisotropy (FA), mean diffusivity (MD) and DTI scalars were evaluated for each patient with differential tractography. Next, group analysis using correlational tractography was performed. Acquired ASL volumes (proton-weighted, M0, and perfusion-weighted) were used to obtain the Cerebral Blood Flow (CBF) maps. These were analysed using SPM12 performing a paired t-test to obtain changes in perfusion after rehabilitation. ANCOVA technique was used to neglect basal perfusion in subjects.

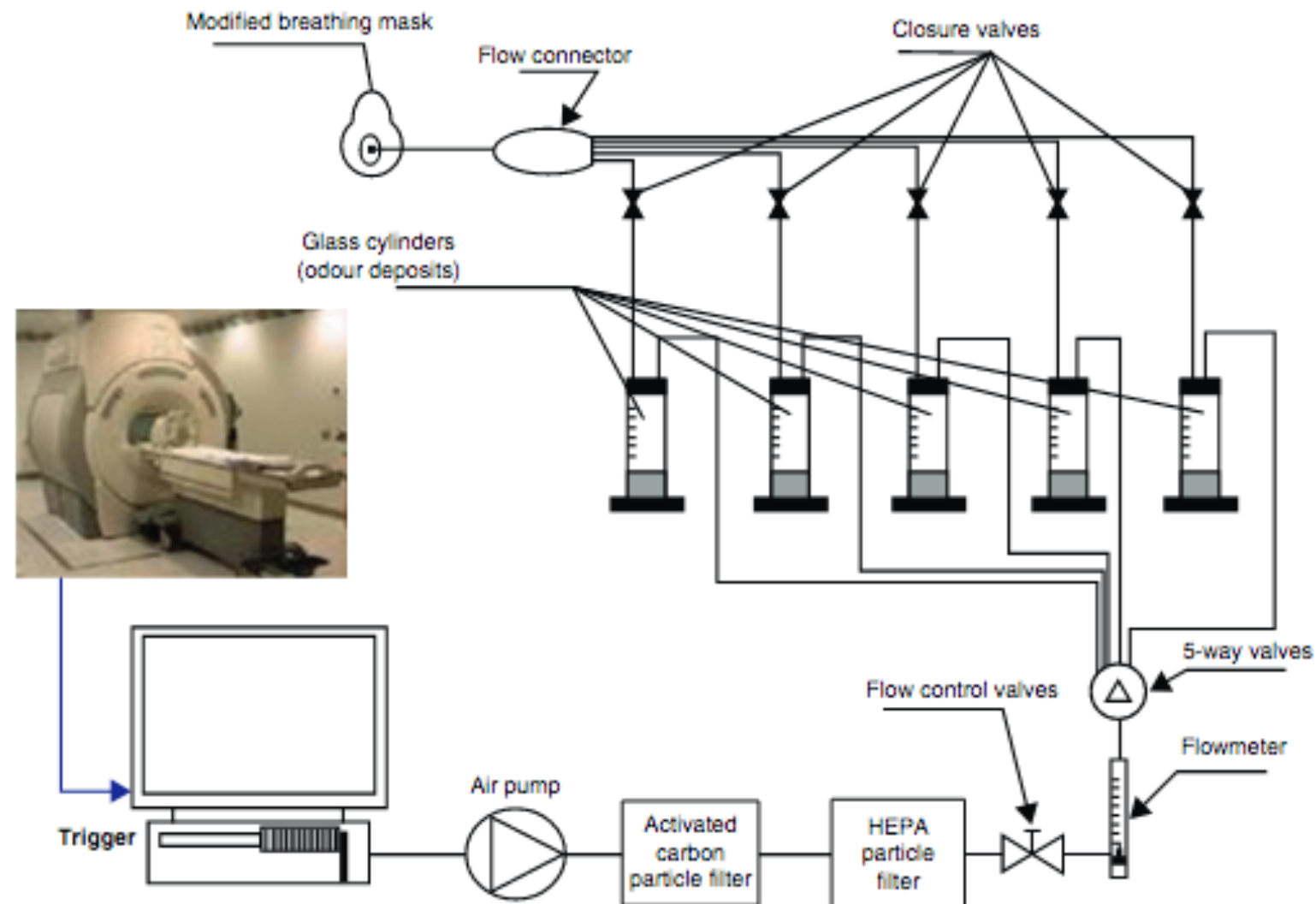


Fig. 1 Scheme of the fully synchronized olfactometer device

Results

VBM analyses ($p_{unc}<0.05$) show increased volumes in the cerebellum, temporal lobe, parahippocampal region and cuneus at therapy onset, which may be related to inflammation due to virus infection (Fig. 2) . Brain regions involved in olfactory rehabilitation such as the fusiform gyrus, caudate and orbital gyrus showed a volume increase ($p_{unc}<0.05$) (Fig. 3). Olfactory rehabilitation therapy has an impact on the activation ($p_{unc}<0.05$) of several brain areas after odorant delivery, including parahippocampal gyrus, frontal gyrus, amygdala, caudate, putamen and insula (Fig. 4). The correlational tractography analysis (T-score = 2) showed increased FA (Fig. 5) in the forceps minor, which allows for interhemispheric connection in the frontal lobe. Correlational tractography also showed decreased RD (T=2) (Fig. 6) in the forceps major, which allows for interhemispheric connection in the occipital and posterior parietal lobes. Perfusion analyses shows perfusion increment in the precuneus, cuneus, right dorsolateral frontal cortex, left middle frontal/precentral gyrus, parahippocampal/fusiform gyrus, insular cortex/putamen, orbitofrontal cortex and cerebellum (p -value=0.05) (Fig. 7).

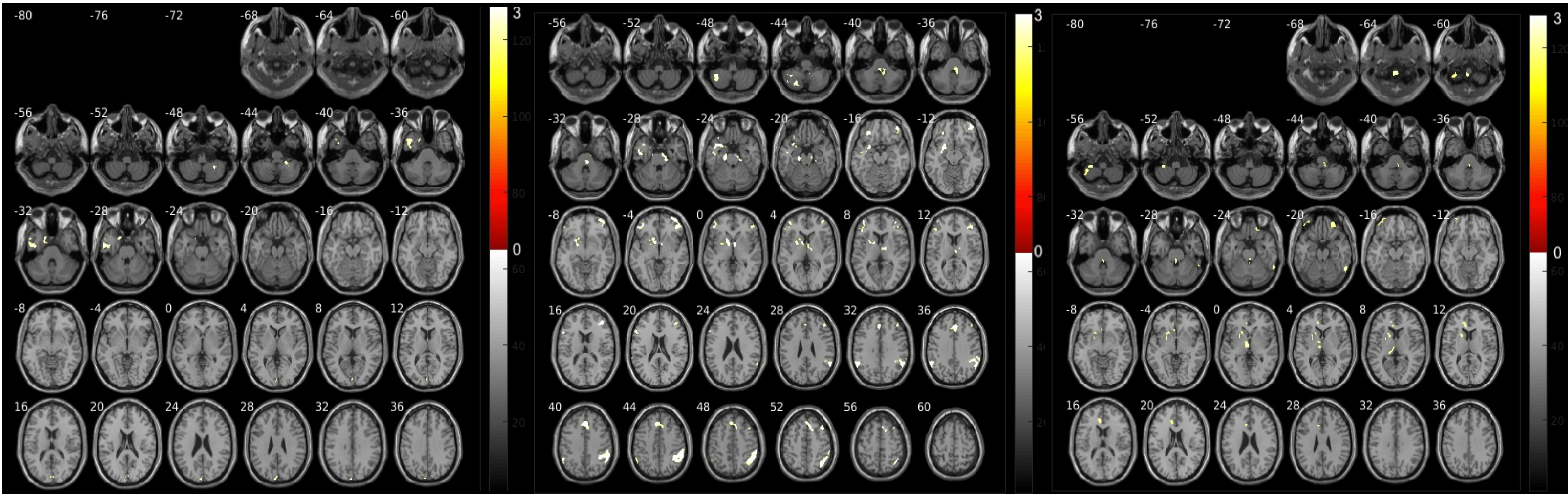


Fig. 2 Decreased volume after rehabilitation therapy ($p_{unc}<0.05$)

Fig. 3 Increased volume after rehabilitation therapy ($p_{unc}<0.05$)

Fig. 4 Increased fMRI activity after rehabilitation therapy ($p_{unc}<0.05$)

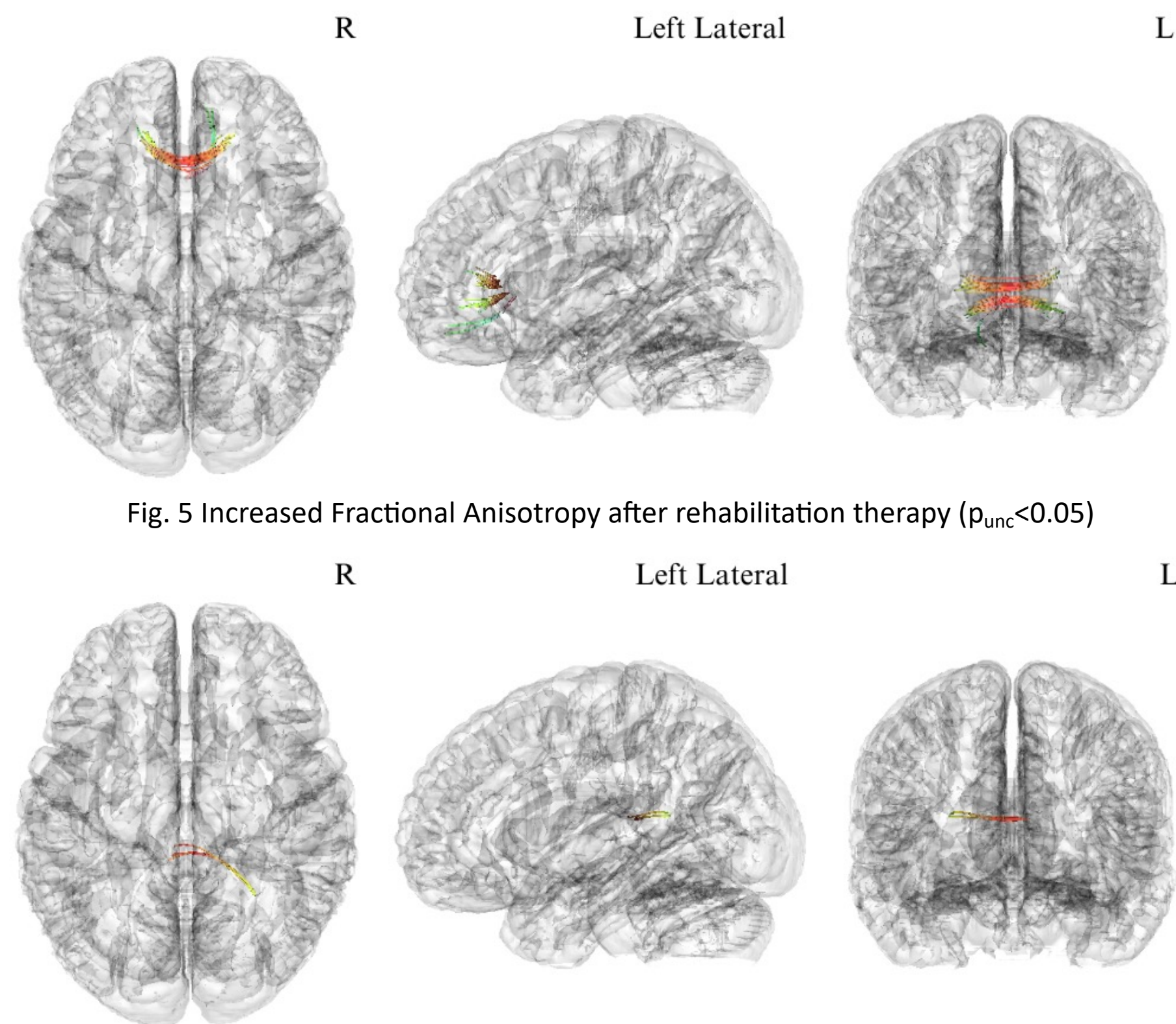


Fig. 5 Increased Fractional Anisotropy after rehabilitation therapy ($p_{unc}<0.05$)

Fig.6 Increased Radial Diffusivity after rehabilitation therapy ($p_{unc}<0.05$)

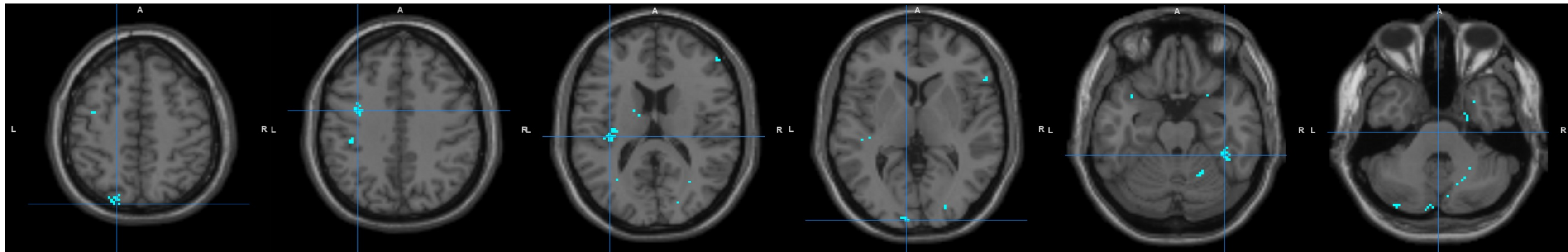


Fig. 7 Increased CBF volume after rehabilitation therapy ($p_{unc}<0.05$)

Discussion

Changes seen in volumetry could be caused by inflammatory processes that decline during the time of rehabilitation with the counterpart of increased volumes in the areas targeted in the olfactory rehabilitation. Functional changes are correlated with patient qualitative scores on rough odour discrimination and threshold activation. Diffusion measures indicate improved white matter connectivity in pathways related to brain regions affected by COVID-19 in anosmic patients, consistent with previous studies⁵. Perfusion increases in the cerebellum, dorsolateral prefrontal cortex⁶, and occipital lobe might indicate the efficacy of olfactory therapy. The alterations found in the different modalities comparisons relate to the cognitive pathways addressed by the rehabilitation therapy such as working memory, decision-making, and emotional or semantic processing, underwent a process of remodelling according to the findings.

Conclusion

These results suggest that olfactory rehabilitation may have a beneficial effect on the neural plasticity of the olfactory system and improve the quality of life of patients with COVID-19-induced anosmia and fMRI, VBM, DTI and ASL assessment provide the necessary tools to perform a comprehensive and complete analysis of the effects of the therapy.

References

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